Examining the California Air Resource Board Greenhouse Gas Reduction Quantification Methodology

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Introduction

The California Air Resources Board (CARB) offers incentives to apply fuel treatments with the intention of ultimately reducing greenhouse gas emissions in the state of California. CARB quantifies these potential reductions in emissions through the Fire and Fuels Extension (FFE) to the Forest Service's Forest Vegetation Simulator (FVS). Fuel reduction treatments are therefore judged by CARB based on the treatment's ability to ultimately reduce emissions. This report attempts to use the FVS outputs to quantify potential carbon reductions for various fuel reduction treatments, including prescribed burn, mechanical thinning, and a combination of the two. Data was collected from two studies conducted at the same plots at the University of California Blodgett Forest Research Station. After fuel treatments were applied to specified stands, researchers gathered biomass data 1-year post-treatment (2001) and then again 7-years post-treatment (2008). We used the 2001 data to simulate above-ground carbon (biomass) accumulation after a fuel treatment over a period of 20 years using FVS. These projections were then compared to the 2008 data to verify and validate FVS' projections. By comparing these projections, we suggest the need for better modelling capabilities or better documentation on proper FVS inputting to ensure optimal calculation of net GHG benefits.

Methods

First, we fully carried out CARB methodology and used the CARB net GHG benefit calculator using pre-treatment data from Blodgett Research Forest (California Air Resources Board, 2018). We then analyzed the process at several stages. To evaluate the most sensitive parameters within this calculation of GHG benefits, we varied our initial inputs to look at the relative sensitivity of GHG benefits to changes in individual parameters. By assessing sensitivity of parameters, we were able to determine which parameters had the greatest potential to impact

the greenhouse gas reduction benefit calculated by a grant applicant. Next, we ran FVS simulations using data gathered from the two studies conducted at Blodgett Research Forest (Stephens, S. L., & Moghaddas, J. J., 2005), and examined how these simulated values compared to real carbon measured in the field. After comparing FVS outputs to real data, we assessed how well we were able to input fuel treatments into the model and how growth after treatment compared between simulations and reality. This led us to recommendations concerning CARB's guidelines for use of FVS.

Description of Data

Data were collected from the University of California Blodgett Forest Research Station (Blodgett Forest), a Sierra Nevada mixed conifer forest in the north-central Sierra Nevada encompassing an area of 1,780 ha. Dominant tree species include sugar pine (*Pinus lambertinana*), ponderosa pine (*Pinus ponderosa*), white fir, incense-cedar, Douglas-fir (*Pseudotsuga menziesii*), California black oak (*Quercus kelloggii*), tan oak (*Lithocarpus densiflorus*), bush chinkapin (*Chrysolepis sempervirens*), and Pacific madrone (*Arbutus menziezii*). Soils at Blodgett Forest are well-developed, with common soil depths of 85 to 115 cm. Slopes average less than 30%. The climate is Mediterranean with a summer drought period that extends into the fall. Most precipitation falls during the winter and spring, averaging 160 cm. Average temperatures in January range from 32℉ to 46.4℉, while summer months tend to range from 50℉ to 84.2℉. Prior to fire suppression policy, fire was common in the region, with median composite fire intervals at the 9 - 15ha spatial scale of 4.7 years and a fire interval range of 28 years (Stephens, S. L., & Moghaddas, J. J., 2005).

Greenhouse Gas Reduction Potential at Blodgett Forest

When we followed the CARB protocol for greenhouse gas accounting, using only the pre-treatment information at the three treatment sites at Blodgett Forest, we found negative greenhouse gas reduction values (e.g. net emissions) for every treatment. Burning was the least harmful fuel treatment in terms of greenhouse gas emissions. The inputs used, found by following CARB's protocol, and outputs found were as follows:

	Fuels Reduction Activity 1
Area within the treatment boundary (acres)	**115.1**
Annual probability of fire occurrence (%)	0.41%
Effective period for fuels reduction treatment (Years)	20
Carbon within the treatment boundary at the end of the project with fuels reduction treatment but without fire disturbance (MT C)	1,697
Carbon within the treatment boundary at the end of the project with fuels reduction treatment and with fire disturbance (MT C)	1,525
Carbon within the treatment boundary at the end of the project without fuels reduction treatment and without fire disturbance (MT C)	3,255
Carbon within the treatment boundary at the end of the project without fuels reduction treatment but with fire disturbance (MT C)	2,128
Biomass removed via mechanical treatments (BDT)	931

GHG benefit from fuels reduction activity 1 (MT CO_2e)	-5,497
On-site carbon storage and project emissions in fuels reduction project scenario (MT CO_2e)	6,122
On-site carbon storage in baseline scenario (MT CO_2e)	11,620

Table 1. CARB Greenhouse Gas Reduction Potential calculator with FVS-FFE-simulated carbon values for mechanical treatment area at Blodgett Forest.

	Fuels Reduction Activity 2
Area within the treatment boundary (acres)	**95.27**
Annual probability of fire occurrence (%)	0.41%
Effective period for fuels reduction treatment (Years)	20
Carbon within the treatment boundary at the end of the project with fuels reduction treatment but without fire disturbance (MT C)	1,443
Carbon within the treatment boundary at the end of the project with fuels reduction treatment and with fire disturbance (MT C)	1,332
Carbon within the treatment boundary at the end of the project without fuels reduction treatment and without fire disturbance (MT C)	2,576
Carbon within the treatment boundary at the end of the project without fuels reduction treatment but with fire disturbance (MT C)	1,731
Biomass removed via mechanical treatments (BDT)	814

GHG benefit from fuels reduction activity 2 (MT CO_2e)	-3,994
On-site carbon storage and project emissions in fuels reduction project scenario (MT CO_2e)	5,215
On-site carbon storage in baseline scenario (MT CO_2e)	9,209

Table 2. CARB Greenhouse Gas Reduction Potential calculator with FVS-FFE-simulated carbon values for mechanical and prescribed fire treatment area at Blodgett Forest.

	Fuels Reduction Activity 3
Area within the treatment boundary (acres)	**99.43**
Annual probability of fire occurrence (%)	0.41%
Effective period for fuels reduction treatment (Years)	20
Carbon within the treatment boundary at the end of the project with fuels reduction treatment but without fire disturbance (MT C)	2,513
Carbon within the treatment boundary at the end of the project with fuels reduction treatment and with fire disturbance (MT C)	2,060
Carbon within the treatment boundary at the end of the project without fuels reduction treatment and without fire disturbance (MT C)	2,700
Carbon within the treatment boundary at the end of the project without fuels reduction treatment but with fire disturbance (MT C)	1,562
Biomass removed via mechanical treatments (BDT)	0

GHG benefit from fuels reduction activity 3 (MT CO_2e)	-488
On-site carbon storage and project emissions in fuels reduction project scenario (MT CO_2e)	9,092
On-site carbon storage in baseline scenario (MT CO_2e)	9,580

Table 3. CARB Greenhouse Gas Reduction Potential calculator with FVS-FFE-simulated carbon values for prescribed burn treatment area at Blodgett Forest.

Annual probability of fire occurrence was determined using the Fire Probability for Carbon Accounting tool (CalFire, http://egis.fire.ca.gov/FireProbability/). Effective period of treatment was chosen to be 20 years, near the upper end of feasible fuel reduction effectiveness. In practice, this figure is normally chosen in consultation with a Registered Professional Forester. We selected this length to find a GHG reduction benefit near the upper bounds of potential benefits.

If we had run these estimates as grant applicants, we may be dissuaded from applying in that our fuel treatment project had negative greenhouse gas reduction potential. However, we were interested in determining how sensitive this calculator was to variable inputs. By comparing relative sensitivity of inputs, we can understand which inputs impact GHG benefits the most.

Sensitivity Analysis of CARB GHG Reduction Potential Calculator to Inputs

The CARB uses several parameters (as seen in the *Table 4* below) from FVS outputs to determine the GHG benefits of a fuel reduction treatment. Fire probability and effective period of fuel treatment are CARB-specific parameters that determine the net benefits of a fuel treatment, while the other entries are gathered from specific FVS outputs. By comparing the sensitivity of these various parameters, we suggest that correct FVS outputs are more important measurements in calculating net GHG benefits.

To conduct this analysis, we used FVS outputs to get a baseline net GHG benefit using CARB's calculator ("Fuels Reduction" 2018). Then, holding all other parameters constant, we changed an individual parameter by 20% to calculate its effect on GHG benefits. The effects to GHG benefits with this 20% change to each individual parameter can be seen in the first column of the table below. In the subsequent column, we calculated the percent change in net GHG benefits from the baseline GHG benefits after each 20% change to an individual parameter.

As seen in *Table 4*, most of these parameters are insensitive to change from our baseline and thus do not contribute much to fluctuations in GHG benefits. However, as should be expected, modifying the carbon remaining in the treatment area without disturbance impacts the

greenhouse gas reduction result significantly (since it implies a that a different amount of carbon is lost to wildfire, which is the emission that the calculator tries to quantify in the first place). For a grant applicant, these values are determined purely through FVS simulation, so it is important to determine whether or not we should trust FVS results.

***Baseline GHG Benefits = -3,994**	GHG Benefit after 20% change in parameter	Percent change from baseline after 20% change in parameter
Area within the treatment boundary (acres)	-3994	0.0
Annual probability of fire occurrence (%)	-3954	1.0
Effective period for fuels reduction treatment (Years)	-3954	1.0
Carbon within the treatment boundary at the end of the project with fuels reduction treatment but without fire disturbance (MT C)	-3019	24.4
Carbon within the treatment boundary at the end of the project with fuels reduction treatment and with fire disturbance (MT C)	-3917	1.9
Carbon within the treatment boundary at the end of the project without fuels reduction treatment and without fire disturbance (MT C)	-5735	43.6
Carbon within the treatment boundary at the end of the project without fuels reduction treatment but with fire disturbance (MT C)	-4095	2.5
Biomass removed via mechanical treatments (BDT)	-4004	0.2

Table 4. Percent change in GHG benefit as a result of 20% change in FVS input parameters.

Examining FVS Carbon Simulations

This sensitivity analysis suggests that the success of the CARB calculator depends mostly on the accuracy of FVS simulations at representing the carbon content of stands. We ran FVS on actual stands at Blodgett Forest for which biomass was calculated (using the same allometric relationships as FVS) and compared simulated results to real results collected from Blodgett Forest (Stephens, Scott L., et al., 2009). Simulations were run using the FVS-Suppose interface with the Western Sierra Nevada, CA variant. The inputs used to run these simulations were

directly based on descriptions of treatments in the Stephens research report, and all treatments were applied in 2001 (Stephens, S. L., & Moghaddas, J. J., 2005):

Prescribed burn:

Treatment was applied in the fall, 50 °F, with wind speeds of 4mph, and fuels were moderately dry. It was assumed that 100% of the treatment area was burned.

Mechanical thinning:

First, a thin from below was administered with a residual density of 30 m^2/acre. Trees of all DBH sizes were included and we assumed no small trees were left, nor any slash left. Next, a mastication was applied to the stands. Trees of DBH of 2 to 25 cm were masticated, and 90% of the trees were masticated. The proportion of surface fuel composed of masticated pieces was kept at the default value of 0.7. It was assumed that both of these activities were applied in the same year, with no gap between applications.

Mechanical thinning and prescribed burn:

Mechanical thinning treatment was applied exactly as described above. The area was thinned from below first, followed by a mastication of the area given the inputs as described above. The area was then burned by the same specifications as in the prescribed fire treatment. It was assumed that both treatments were applied to the area in the same year, with no gap between applications.

Wildfire simulation:

Wildfire was simulated using the SimFire keyword in the FFE. Inputs were kept at default settings, with wildfire occurring in the early spring, 70 °F, wind speeds of 20mph, while fuels were very dry. It was assumed that 70% of the treatment area was burned and that wildfire disturbance occurred 10 years post-treatment (2010) for all plots that experienced wildfire. By experiencing a wildfire disturbance 10 years after treatment, stands accumulated sufficient biomass but stand structure was still impacted by the original fuel treatment received at the onset of the project.

After simulations were complete, the outputted carbon report was used to compare results across treatments and track the biomass accumulation of above-ground carbon. Below are the results of all FVS simulations, separated by treatment type (control, prescribed burn, mechanical thinning, and mechanical thinning plus prescribed fire).

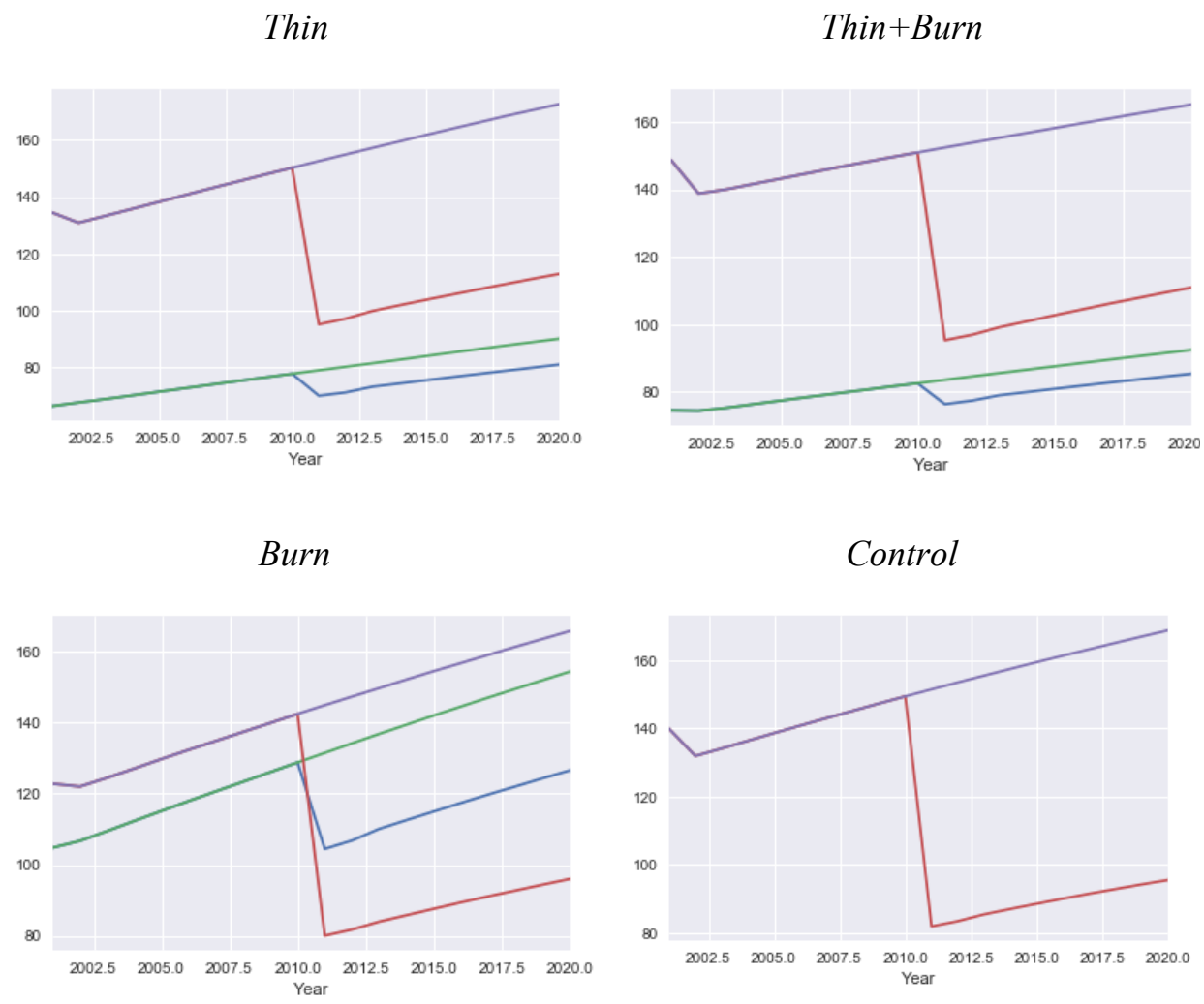

Figure 1. Sum of total stand carbon across stands in each treatment area, with 20-year trajectories for carbon with and without treatment application and with or without a fire disturbance. Y-axis unit is Mean Mg C/ha across stands of each treatment type.

FVS demonstrated linear growth of total stand carbon, with dips of varying magnitudes in those simulations that experienced a wildfire in 2010. These trajectories show the lack of nuance in the FVS model - growth is strictly linear, without changes in rate over time or after disturbance. We used real data collected in 2001 and 2008 to validate these FVS carbon estimates. This was done by comparing real data mean and standard error calculations to FVS outputted means, as seen below.

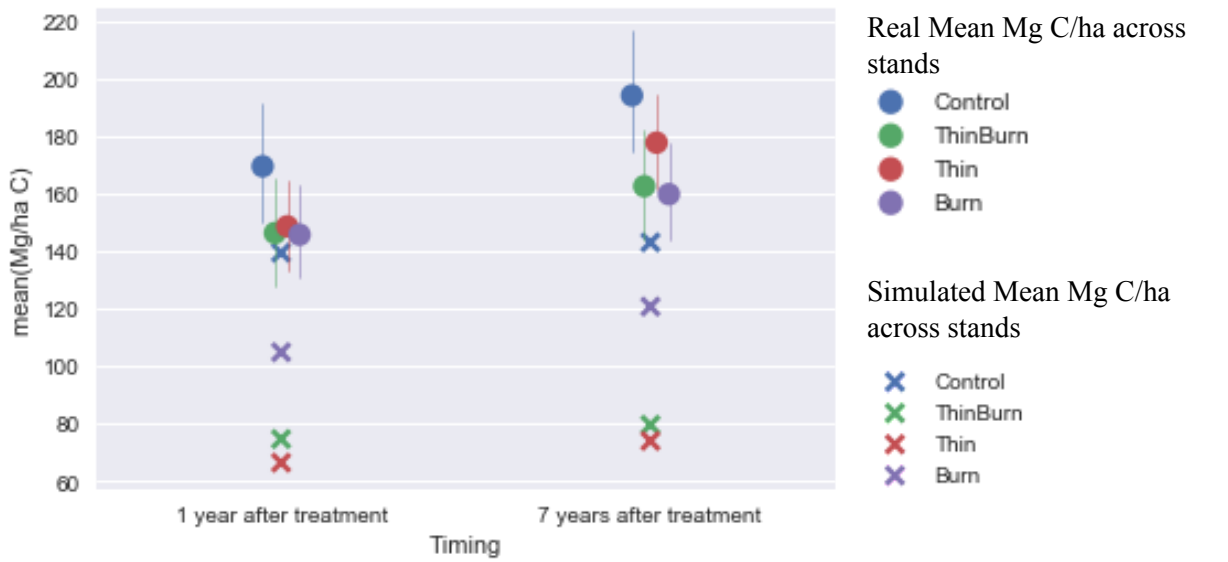

Figure 2. Mean Mg C/ha (from real data) in each treatment type 1 year and 7 years after treatment, as well as mean Mg C/ha in simulations. Error bars represent 1 standard error.

Figure 2 shows real average aboveground carbon (± 1 SE) in each treatment area 1 year and 7 years after treatment. Although all treatments had lower carbon stock in FVS simulations than was measured in the field, the two treatments that include a mechanical thinning were simulated to have nearly two times less carbon than what was actually measured in the treatment areas. One explanation of why these treatments in particular were so egregious is that the mechanical treatment specified in FVS in particular removed more biomass relative to what was actually removed in the field. In control, thin, and thin-burn treatments, the disparity between what was simulated and what was measured in the field increased over time, meaning that the

20-year carbon values ultimately used as inputs to the CARB calculator may be even more distant from reality.

Timing	Treat	Real Mg/ha C, (mean ± 1 SE)	Simulated Mean Mg/ha C
Post1	Burn	140.89 ±8.30, n = 60	104.81
Post7	Burn	149.42 ±8.47, n = 60	120.68
Post7/Post1 Real:	**1.06**	**Post7/Post1 Simulated:**	**1.15**
Post1	Control	164.43 ±9.99, n = 60	139.90
Post7	Control	189.87 ±11.10, n = 58	143.03
Post7/Post1 Real:	**1.15**	**Post7/Post1 Simulated:**	**1.02**
Post1	Thin	141.37 ±7.31, n = 60	66.35
Post7	Thin	171.98 ±8.65, n = 59	73.93
Post7/Post1 Real:	**1.22**	**Post7/Post1 Simulated:**	**1.11**
Post1	ThinBurn	139.89 ±9.12, n = 60	74.58
Post7	ThinBurn	151.04 ±9.64, n = 59	79.53
Post7/Post1 Real:	**1.08**	**Post7/Post1 Simulated:**	**1.07**

Table 5. Means and standard errors of stand aboveground live carbon, and simulated values for aboveground live carbon. The ratio between the year 7 value and year 1 value is shown for both real and simulated data to compare by what factor the carbon held in the treatment areas increased between the real and simulated data.

It is clear from Figure 2 that FVS failed to represent the 1-year data, but given that the difference between the real and simulated values in year 7 was even greater than in year 1, this means that FVS additionally failed to model 6 years of recovery correctly (showing less growth than really occurred). Table 5 shows the ratio between the year 7 and year 1 carbon stock in the simulated and real data - in all but the burn treatment, the simulated ratio of growth is lower than the real factor by which carbon increased. This shows that in addition to starting with incorrect values, FVS does not model growth over 6 years properly.

Given that our inputs to FVS were directly taken from the stated objectives of the thinning treatments in the Stephens and Moghaddas paper (Stephens, S. L., & Moghaddas, J. J.,

2005), this points to a source of uncertainty in later CARB greenhouse gas calculations. These differences could be exacerbated over longer treatment lengths.

Impact of Use of Better Inputs to CARB Calculator

To investigate the effect of more realistic carbon inputs, we tried using real 2007 data collected from the later Stephens paper (Stephens, Scott L., et al., 2009) as rough, but hopefully more realistic, estimates of true greenhouse gas reduction potential rather than using simulated results. We used the following values and relationships:

Carbon inside the treatment boundary, with fuels reduction treatment and without fire:

Real 7-year total carbon in the "thin" treatment, times the ratio of simulated 20-year carbon to simulated 7-year carbon to account for further growth. Although we do not believe that the FVS growth model is realistic (as described above), we still needed to incorporate more than 7 years of growth into our calculations.

Carbon inside the treatment boundary, with fuels reduction treatment and with fire:

The real 2007 total carbon in the "thin" treatment, multiplied by the ratio of carbon in simulated treatments with fire to the carbon in simulated treatments without fire (assuming that fire would consume a similar proportion of standing biomass in our reality as in our simulations), times the ratio of simulated 20-year carbon to simulated 7-year carbon to account for further growth.

Carbon inside the treatment boundary, without fuels reduction treatment and without fire:

Real 2007 total carbon in the "control" treatment, multiplied by the ratio of carbon in the thin treatment to the carbon in the control treatment before any treatment occurred (to account for different initial biomass/area of the different treatment areas), times the ratio of simulated 20-year carbon to simulated 7-year carbon to account for further growth.

Carbon inside the treatment boundary, with fuels reduction treatment and with fire:

Our estimate of carbon without fuels treatment (as above, real 2007 total carbon in the "control" treatment * pre-treatment "thin" carbon/ "control" carbon), again multiplied by the ratio of carbon in simulated treatments with fire to the carbon in simulated treatments without

fire, times the ratio of simulated 20-year carbon to simulated 7-year carbon to account for further growth.

Biomass removed by mechanical treatment:

Half of the difference between the carbon in the treatment area with and without the treatment (this is a somewhat arbitrary choice, but is similar to the values in our simulations).

	Fuels Reduction Activity 4
Area within the treatment boundary (acres)	115.1
Annual probability of fire occurrence (%)	0.41%
Effective period for fuels reduction treatment (Years)	20
Carbon within the treatment boundary at the end of the project with fuels reduction treatment but without fire disturbance (MT C)	9,612
Carbon within the treatment boundary at the end of the project with fuels reduction treatment and with fire disturbance (MT C)	8,559
Carbon within the treatment boundary at the end of the project without fuels reduction treatment and without fire disturbance (MT C)	10,777
Carbon within the treatment boundary at the end of the project without fuels reduction treatment but with fire disturbance (MT C)	9,596
Biomass removed via mechanical treatments (BDT)	582

GHG benefit from fuels reduction activity 4 (MT CO_2e)	-4,273
On-site carbon storage and project emissions in fuels reduction project scenario (MT CO_2e)	34,936
On-site carbon storage in baseline scenario (MT CO_2e)	39,210

Figure 3. California ARB Greenhouse Gas Reduction Potential calculator with estimates based on real carbon values for mechanical thinning treatment area at Blodgett Forest.

With these modified inputs, the greenhouse gas reduction potential was still negative for our most-emitting treatment, but slightly less so (-4,273 MT CO_2e instead of -5,497 MT CO_2e). In cases where a manager's treatment might have a very slight greenhouse gas benefit, this margin of error could result in a negative calculation that discourages a manager from applying for CARB funding.

Recommendations

From the sensitivity analysis, we see that carbon outputs from FVS are the most sensitive parameter in determining GHG benefits. However, based on our comparisons to real data, we observed that FVS projections are not correctly representing what is actually seen in the field, both at the level of treatment application and at the level of post-treatment recovery. Therefore, CARB should consider providing better documentation on proper use of FVS for modelling

initial treatments, and reconsider the validity of the model for describing growth. The GHG reduction calculator is most sensitive to relative carbon stock between the different scenarios.

For example, the mechanical thinning treatment had an objective of leaving 28-34 m^2/ha of basal area during the crown thin operation. In our initial simulations, we assumed that a grant applicant would pick a value somewhere in the middle of their target range in order to apply to the program, and chose 30 m^2/ha for our simulation. Re-running this simulation with an input at the top of the objective range (34 m^2/ha basal area remaining after thinning) gave us values very slightly closer to reality: (66.35 Mg C/ha in 2001 and 73.92 Mg C/ha in 2008 in our original simulation were updated to 67.30 and 75.01 respectively, still far below the actual values of 141.37 and 171.98 Mg/C).

Modifications to other FVS inputs could have a cumulative effect that, when coupled with poor FVS projection of data, could ultimately determine whether or not GHG benefits are positive. This sensitive relationship and accumulation of applicant and model errors could determine whether or not a manager may receive a grant to complete beneficial fuel treatments.

Grant applicants should also be aware of the danger of inputting management objectives to FVS, rather than what they will reasonably accomplish during actual application of the fuel treatment. In the current guidance from CARB, applicants are not cautioned to take this into account. To mitigate this error, CARB should provide clearer explanations of inputs, as well as provide examples of simulations for a set of standard fuel management practices. This guidance may help managers to use more realistic numbers when calculating GHG benefits, and therefore may result in better estimations of benefits. With better estimation, more fuel reduction treatments that are beneficial by CARB standards will rightfully receive grants rather than be turned away for mistakenly using incorrect input values. While this does not solve the issue of model-prone errors of poor projection of data, minimizing manager-prone errors is a step in the right direction to optimize funding for beneficial fuel reduction treatments.

Conclusion

The Air Resources Board provides a set of guidelines and tools to estimate the greenhouse gas reduction potential of fuel reduction practices. Our investigation of the

methodology was greatly limited by the availability of data - we only evaluated the effectiveness of FVS simulations for four areas, and thus were unable to statistically quantify whether simulated outputs differed from realistic growth. As such, this analysis is mostly a qualitative investigation of the steps in the CARB methodology in which a user could cause variability in results. However, through this qualitative investigation, we find that these protocols are (understandably) sensitive to the difference between the carbon stored in fire-affected ecosystems with and without fuels treatment, so the actual magnitude of the greenhouse gas reduction (or emission) potential is dependent on users first creating realistic values for the carbon stored in their areas of interest. Providing more guidance on the use of FVS-FFE, including reasonable inputs and caveats as to the validity of the model, should be a priority of the Greenhouse Gas Reduction Fund program.

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